```sql
--use cosmos_sneakpeek;
use Cosmos;


DROP TABLE IF EXISTS #Diabetes ;--6561
SELECT DISTINCT DiagnosisKey,
                TerminologyConceptKey,
                [Value],
                DisplayString, -- dd.Name ,NameAndCode,GroupedNameAndCode
                CASE
                    WHEN DisplayString LIKE '%right eye%' THEN 'RT'
                    WHEN DisplayString LIKE '%left eye%' THEN 'LT'
                    WHEN DisplayString LIKE '%bilateral%' THEN 'BILAT'
                    WHEN DisplayString LIKE '%unspecified eye%' THEN 'UNSPEC'
                    ELSE 'UNSPEC' -- Use 999 or another code for cases with no laterality info
                END AS Laterality,
                CASE
                    WHEN DisplayString LIKE '%with macular edema%' THEN 'Yes'
                    WHEN DisplayString LIKE '%without macular edema%' THEN 'No'
                    ELSE '999'
                END AS Macular_Edema_status,
                CASE
                   WHEN DisplayString LIKE '%Type 1%' THEN '1'
                   WHEN DisplayString LIKE '%Type 2%' THEN '2'
                   ELSE '999'
                END AS DM_type



INTO #Diabetes
                FROM DiagnosisTerminologyDim
                --left join DiagnosisDim dd
                -- on dd.DiagnosisKey=DiagnosisTerminologyDim.DiagnosisKey
                WHERE [Type]=N'ICD-10-CM'
                    AND (
                        (([Value] LIKE 'E10.3%' AND [Value] != 'E10.3') AND [Value] NOT LIKE 'E10.3[6-9]%') OR
                        (([Value] LIKE 'E11.3%' AND [Value] != 'E11.3') AND [Value] NOT LIKE 'E11.3[6-9]%') OR
                        [Value] LIKE 'E08.9%' OR
                        [Value] LIKE 'E09.9%' OR
                        [Value] LIKE 'E10.9%' OR
                        [Value] LIKE 'E11.9%'
                )   AND [Value] NOT LIKE 'E08.31%'
                    AND [Value] NOT LIKE 'E09.31%'
                    AND [Value] NOT LIKE 'E10.31%'
                    AND [Value] NOT LIKE 'E11.31%'

                    ;



DROP TABLE IF EXISTS #Diabetes_unique; --138
select distinct [Value],
                DisplayString,
                CASE SUBSTRING([Value], 6, 1)
                WHEN '2' THEN 'Mild NPDR'
                WHEN '3' THEN 'Moderate NPDR'
                WHEN '4' THEN 'Severe NPDR'
                WHEN '5' THEN 'PDR'
                ELSE 'No Retinopathy'
            END AS retinopathy_severity,

                CASE SUBSTRING([Value], 6, 1)
                WHEN '2' THEN '1'
                WHEN '3' THEN '2'
                WHEN '4' THEN '3'
                WHEN '5' THEN '4'
                ELSE '0'
            END AS SeverityRank,
                case when [Value] in ('E10.9','E11.9','E08.9','E09.9') Then 'Bilateral'
                else Laterality
                end as Laterality,
                Macular_Edema_status,
                DM_type
into #Diabetes_unique
From #Diabetes;



DROP TABLE IF EXISTS #DR_severe;
select d.DiagnosisKey,d.TerminologyConceptKey,d.[Value],d.DisplayString,d.Macular_Edema_status,d.DM_type,
       du.retinopathy_severity, du.SeverityRank,du.Laterality
into #DR_severe
from #Diabetes d
join #Diabetes_unique du
    on du.[Value]=d.[Value]
where du.SeverityRank >=3
    ;



------------------------------



DROP TABLE IF EXISTS #complications ;--6561
SELECT DISTINCT DiagnosisKey,
                TerminologyConceptKey,
                [Value],
                DisplayString, -- dd.Name ,NameAndCode,GroupedNameAndCode
                CASE
                    WHEN DisplayString LIKE '%right eye%' THEN 'RT'
                    WHEN DisplayString LIKE '%left eye%' THEN 'LT'
                    WHEN DisplayString LIKE '%bilateral%' THEN 'BILAT'
                    WHEN DisplayString LIKE '%unspecified eye%' THEN 'UNSPEC'
                    ELSE 'UNSPEC' -- Use 999 or another code for cases with no laterality info
                END AS Laterality,

            CASE
                WHEN [Value] like N'H43.1%' Then 'Vitreous Hemorrhage'
                WHEN [Value] like N'H40.%' THEN 'Neovascular Glaucoma'
                ELSE 'Traction Retinal Detachment'
            END as Complication_grp
INTO #complications
                FROM DiagnosisTerminologyDim
                WHERE Type=N'ICD-10-CM'
                AND ([Value] IN ('H43.10', 'H43.11', 'H43.12', 'H43.13',
                                 'H40.89'
                                --'H40.5',
                                --'H33.4','E08.352', 'E08.353', 'E08.354', 'E09.352', 'E09.353', 'E09.354',
                                --'E10.352', 'E10.353', 'E10.354', 'E11.352', 'E11.353', 'E11.354'
                                )
                                OR [Value] LIKE N'H40.5%'
                                OR [Value] LIKE N'H33.4%'
                                OR [Value] LIKE N'E08.352%'
                                OR [Value] LIKE N'E08.353%'
                                OR [Value] LIKE N'E08.354%'
                                OR [Value] LIKE N'E09.352%'
                                OR [Value] LIKE N'E09.353%'
                                OR [Value] LIKE N'E09.354%'
                                OR [Value] LIKE N'E10.352%'
                                OR [Value] LIKE N'E10.353%'
                                OR [Value] LIKE N'E10.354%'
                                OR [Value] LIKE N'E11.352%'
                                OR [Value] LIKE N'E11.353%'
                                OR [Value] LIKE N'E11.354%'

                    );




DROP TABLE IF EXISTS #initial_cohort ;
```

```sql
select DiagnosisKey,EncounterKey,PatientDurableKey,AgeKey,StartDateKey,
        [Value],DisplayString,Macular_Edema_status,DM_type,
       retinopathy_severity, SeverityRank,Laterality
into #initial_cohort

from (

select def.DiagnosisKey,def.EncounterKey,def.PatientDurableKey,def.AgeKey,StartDateKey,
        --[Type],[Status],PresentOnAdmission,[EmergencyDepartmentDiagnosis],[Chronic],[IsPrimary],[DocumentedInstant_X]
        d.[Value],d.DisplayString,d.Macular_Edema_status,d.DM_type,
       d.retinopathy_severity, d.SeverityRank,d.Laterality ,

       ROW_NUMBER() OVER (
      PARTITION BY def.PatientDurableKey, Laterality
      ORDER BY StartDateKey ASC) AS rn

from DiagnosisEventFact def
 INNER JOIN #DR_severe d ON def.DiagnosisKey = d.DiagnosisKey
 left join EncounterFact ef
    on ef.EncounterKey=def.EncounterKey
INNER JOIN DepartmentDim dept
      ON ef.DepartmentKey = dept.DepartmentKey

  WHERE def.PatientDurableKey > 0 and
    StartDateKey>0 and
    def.AgeKey>0 and
    def.[Type] <> N'Problem List' and
    def.IsPrimary=1
    and (
        dept.DepartmentSpecialty like N'%ophth%' or
        dept.DepartmentSpecialty like N'%optom%'
    )

    ) r

where r.rn=1;


--select count_big(*) from #initial_cohort--5581

--select count_big(distinct PatientDurableKey) from #initial_cohort; --3657

--select count_big(distinct concat( PatientDurableKey, Laterality)) from #initial_cohort; --3657


DROP TABLE IF EXISTS #cohort ;
-- 1. Identify patients with the inclusion criteria: Severe NPDR and PDR
WITH DR_Patients AS (

select * from #initial_cohort

),

-- 2. Identify patients to exclude based on "ever" having these complications
Exclusion_Ever AS (
    SELECT pt.PatientDurableKey,pt.Laterality,pt.StartDateKey,comp.Laterality as comp_lat, def.StartDateKey as comp_date

    FROM (SELECT PatientDurableKey, Laterality, StartDateKey from DR_Patients) pt
    inner join DiagnosisEventFact def
      on def.PatientDurableKey=pt.PatientDurableKey
    inner join #complications comp
      on comp.DiagnosisKey=def.DiagnosisKey


    WHERE
      def.StartDateKey>0 and def.[Type]<>N'Problem List'
      and def.StartDateKey<=pt.StartDateKey
      AND (
        -- Same eye
        (pt.Laterality = 'LT'  AND comp.Laterality = 'LT')
     OR (pt.Laterality = 'RT'  AND comp.Laterality = 'RT')

        -- Bilateral complication affects either eye
     OR (pt.Laterality = 'LT'  AND comp.Laterality = 'BILAT')
     OR (pt.Laterality = 'RT'  AND comp.Laterality = 'BILAT')

        -- Conservative: unspecified complication could affect either eye
     OR (pt.Laterality = 'LT'  AND comp.Laterality = 'UNSPEC')
     OR (pt.Laterality = 'RT'  AND comp.Laterality = 'UNSPEC')

        -- Bilateral cohort eye
     OR (pt.Laterality = 'BILAT' AND comp.Laterality IN ('LT','RT','BILAT','UNSPEC'))

        -- Unspecified cohort eye
     OR (pt.Laterality = 'UNSPEC' AND comp.Laterality IN ('LT','RT','BILAT','UNSPEC'))
   )

),
remaining_pt as (
select c.*


from #initial_cohort c
where not exists
(select 1
  from  Exclusion_Ever e
  where e.PatientDurableKey=c.PatientDurableKey
  and e.Laterality=c.Laterality
  )

),
--select count(distinct  concat( PatientDurableKey, Laterality)) from remaining_pt-- 2632
--select Laterality, count(*) as n from remaining_pt group by Laterality ;-- 2632

-- 4. Calculate total follow-up duration (at least 5 years)
last_enc AS (

select ef.PatientDurableKey, max(dt.DateValue) as last_encounter_date

from (  SELECT
        PatientDurableKey,
        MIN(StartDateKey) AS Earliest_dx
        --MAX(StartDateKey) AS Latest_Record
    FROM remaining_pt
    GROUP BY PatientDurableKey) pt

join EncounterFact ef
    on ef.PatientDurableKey=pt.PatientDurableKey
LEFT JOIN DateDim dt
      ON dt.DateKey=ef.DateKey

group by ef.PatientDurableKey

),
filtered_encs as (

select pt.PatientDurableKey,pt.Laterality, dt.DateValue as index_date,e.last_encounter_date,
         DATEDIFF(DAY, dt.DateValue , e.last_encounter_date) AS FU_days,
         DATEDIFF(YEAR, dt.DateValue , e.last_encounter_date) AS FU_yrs

from remaining_pt pt
left join last_enc e
    on e.PatientDurableKey=pt.PatientDurableKey
LEFT JOIN DateDim dt
      ON dt.DateKey=pt.StartDateKey
where DATEDIFF(YEAR, dt.DateValue , e.last_encounter_date)>=5
)

select DiagnosisKey,EncounterKey,e.PatientDurableKey, --AgeKey,StartDateKey,
        e.index_date,dd.Years as age,
        [Value],c.DisplayString,Macular_Edema_status,DM_type,
       retinopathy_severity, SeverityRank,e.Laterality,

 e.last_encounter_date,e.FU_yrs,e.FU_days
```

```
into #cohort

from remaining_pt c
join filtered_encs  e
 on e.PatientDurableKey=c.PatientDurableKey
 and e.Laterality=c.Laterality
LEFT JOIN DurationDim dd
    ON dd.DurationKey=c.AgeKey
;


--select * FROM #cohort WHERE PatientDurableKey='873013';

--select * FROM #cohort_inferred WHERE PatientDurableKey='873013';

-- select pt.PatientDurableKey, pt.index_date,pt.Laterality,
--        pt2.index_date as index_date2,pt2.Laterality as Lat2
-- from (select * from  #cohort where Laterality='UNSPEC') pt
-- join (select * from  #cohort where Laterality<>'UNSPEC') pt2
-- on pt.PatientDurableKey=pt2.PatientDurableKey and SUBSTRING(pt.[Value], 1, 6) =SUBSTRING(pt2.[Value], 1, 6)
-- ;




--- Now get unspec, infer? then expand bilat.

DROP TABLE IF EXISTS #cohort_inferred ;
with unspec as (
select * from #cohort where Laterality='UNSPEC'
),
matched_pt as (

select u.PatientDurableKey,u.Laterality,u.index_date as index_date,
       dt.DateValue as next_dx_date,
       d.Laterality as inferred_lat,
       u.[Value] as icd10_index, d.[Value] as icd10_inferred,
      DATEDIFF(DAY, u.index_date, dt.DateValue ) as date_diff,

      ROW_NUMBER() OVER (
          PARTITION BY u.PatientDurableKey,u.Laterality
          ORDER BY abs(DATEDIFF(DAY, u.index_date, dt.DateValue )) asc
          ) as rn

from unspec u
join DiagnosisEventFact def
     on def.PatientDurableKey=u.PatientDurableKey
INNER JOIN #DR_severe d ON def.DiagnosisKey = d.DiagnosisKey
LEFT JOIN DateDim dt
     ON dt.DateKey=def.StartDateKey

where
     --dt.DateValue> u.index_date
    d.Laterality<>'UNSPEC'
   and   SUBSTRING(u.[Value], 1, 6) =SUBSTRING(d.[Value], 1, 6)
   ),

inferred_lat as (
select * from matched_pt
where
--PatientDurableKey='229448'
rn =1
and DATEDIFF(DAY, index_date, next_dx_date) <=90
--order by PatientDurableKey,Laterality,next_dx_date
)
select *
into #cohort_inferred
from inferred_lat
;

--select DiagnosisKey,EncounterKey,c.PatientDurableKey, --AgeKey,StartDateKey,

--        case
--          when c.Laterality='UNSPEC' and i.inferred_lat is not null then i.inferred_lat
--          else c.Laterality
--      end as Laterality,

--        c.index_date, age,
--      [Value],DisplayString,Macular_Edema_status,DM_type,
--       retinopathy_severity, SeverityRank,
--       last_encounter_date,FU_yrs,FU_days
--into #cohort_inferred
--from #cohort c
--left join inferred_lat i
--on i.PatientDurableKey=c.PatientDurableKey and i.index_date=c.index_date
;


--select Laterality, count(*) as n from #cohort group by Laterality ;
--UNSPEC    310
--RT    111
--BILAT 354
--LT    91

--select * from #cohort_inferred

--select Laterality, count(*) as n from #cohort_inferred group by Laterality ;

--UNSPEC    279
--RT    113
--BILAT 377
--LT    97


--DROP TABLE IF EXISTS #cohort_test;
--select *
--into #cohort_test
--from #cohort;

ALTER TABLE #cohort
ADD IsInferredLat BIT NULL;



UPDATE  t1
SET t1.Laterality = t2.Inferred_Lat,
t1.IsInferredLat = 1
FROM #cohort AS t1
JOIN #cohort_inferred AS t2
    ON t1.PatientDurableKey = t2.PatientDurableKey
WHERE t1.Laterality = 'UNSPEC'
  --AND t2.Laterality = 'UNSPEC'
  AND t2.Inferred_Lat IS NOT NULL;

  ---------------------------------------------------------------------


 DROP TABLE IF EXISTS #final_cohort;
with expanded_dx as (
  select DiagnosisKey,EncounterKey,PatientDurableKey, 'RT' as Laterality,IsInferredLat,
       index_date, age,
        [Value],DisplayString,Macular_Edema_status,DM_type,
       retinopathy_severity, SeverityRank,
       last_encounter_date,FU_yrs,FU_days

 from #cohort where Laterality = 'BILAT'

 union all
  select DiagnosisKey,EncounterKey,PatientDurableKey,'LT' as Laterality,IsInferredLat,
       index_date, age,
        [Value],DisplayString,Macular_Edema_status,DM_type,
       retinopathy_severity, SeverityRank,
       last_encounter_date,FU_yrs,FU_days
```

```
 from #cohort where Laterality = 'BILAT'

  union all
  select DiagnosisKey,EncounterKey,PatientDurableKey,Laterality,IsInferredLat,
       index_date, age,
        [Value],DisplayString,Macular_Edema_status,DM_type,
       retinopathy_severity, SeverityRank,
       last_encounter_date,FU_yrs,FU_days

 from #cohort where Laterality <> 'BILAT'
),
--select * from expanded_dx
--ranked_lat as (
--select *,
--       case when Laterality in ('LT','RT') then 1
--         else 99
--         end as lat_rank

--from expanded_dx
--),
ranked_dx as (
select *,
       ROW_NUMBER() OVER (PARTITION BY PatientDurableKey,Laterality ORDER BY index_date ASC) AS rn
from expanded_dx
),
eligible_pt as (
select  DiagnosisKey,EncounterKey,PatientDurableKey,Laterality,IsInferredLat,
       index_date, age,
        [Value],DisplayString,Macular_Edema_status,DM_type,
       retinopathy_severity, SeverityRank,
       last_encounter_date,FU_yrs,FU_days

 from #cohort where Laterality = 'BILAT'

 union all
  select DiagnosisKey,EncounterKey,PatientDurableKey,'LT' as Laterality,IsInferredLat,
       index_date, age,
        [Value],DisplayString,Macular_Edema_status,DM_type,
       retinopathy_severity, SeverityRank,
       last_encounter_date,FU_yrs,FU_days

 from #cohort where Laterality = 'BILAT'

  union all
  select DiagnosisKey,EncounterKey,PatientDurableKey,Laterality,IsInferredLat,
       index_date, age,
        [Value],DisplayString,Macular_Edema_status,DM_type,
       retinopathy_severity, SeverityRank,
       last_encounter_date,FU_yrs,FU_days

 from #cohort where Laterality <> 'BILAT'
),
--select * from expanded_dx
--ranked_lat as (
--select *,
--       case when Laterality in ('LT','RT') then 1
--         else 99
--         end as lat_rank

--from expanded_dx
--),
ranked_dx as (
select *,
       ROW_NUMBER() OVER (PARTITION BY PatientDurableKey,Laterality ORDER BY index_date ASC) AS rn
from expanded_dx
),
eligible_pt as (
select  DiagnosisKey,EncounterKey,PatientDurableKey,Laterality,IsInferredLat,
       index_date, age,
        [Value],DisplayString,Macular_Edema_status,DM_type,
       retinopathy_severity, SeverityRank,
       last_encounter_date,FU_yrs,FU_days


from ranked_dx
where rn=1
),
pt_to_remove as (-- those with remaining unspec and could not infer lat withing 90 days
 select pt.PatientDurableKey, pt.index_date,pt.Laterality,
        pt2.index_date as index_date2,pt2.Laterality as Lat2,
        case when pt.index_date<pt2.index_date then 1
        else 0
        end as rvm_flag
 from (select * from  eligible_pt where Laterality='UNSPEC') pt
 join (select * from  eligible_pt where Laterality<>'UNSPEC') pt2
 on pt.PatientDurableKey=pt2.PatientDurableKey
 --and SUBSTRING(pt.[Value], 1, 6) =SUBSTRING(pt2.[Value], 1, 6)
 ),
 reduced as ( -- get all patients with first unspec dx
 select distinct PatientDurableKey
 from pt_to_remove where rvm_flag=1
 )
 select *

into #final_cohort
 from eligible_pt
 where PatientDurableKey not in (select PatientDurableKey from reduced)
  and Laterality <> 'UNSPEC'
 ;


 --select * from #final_cohort




 /*
  ------------------------------ checks ------------------------------
   DROP TABLE IF EXISTS #final_cohort2;
with final_pt as (
select *,
     ROW_NUMBER() OVER (PARTITION BY PatientDurableKey,Laterality ORDER BY index_date ASC) AS rn
from #cohort_test
)
select *
into  #final_cohort2
from final_pt where rn=1
;


with final_pt as (
select *,
     ROW_NUMBER() OVER (PARTITION BY PatientDurableKey,Laterality ORDER BY index_date ASC) AS rn
from #cohort_test
)
select *
from final_pt
order by PatientDurableKey,Laterality
;


 select count(*) from #final_cohort; -- 682
 select count(distinct PatientDurableKey) from #final_cohort; --372
 SELECT COUNT(DISTINCT concat(PatientDurableKey,'-',Laterality))  from #final_cohort;--682


 select Laterality, count(*) from #final_cohort group by Laterality;
LT  337
RT  345



 select count(*) from #final_cohort2; -- 1027
 select count(distinct PatientDurableKey) from #final_cohort2; --619
 SELECT COUNT(DISTINCT concat(PatientDurableKey,'-',Laterality))  from #final_cohort2;--1027
```

```
 select Laterality, count(*) from #final_cohort2 group by Laterality;
UNSPEC  204
RT  419
LT  404

 --todo: remove unspec/infer from first dx with lat. ie patient has lat and unspec.

 */
```